Dynamic Planet Practice Test CSE 2025

Union County Vocational Technical Schools (NJ)

Multiple Choice Question Answer Key:

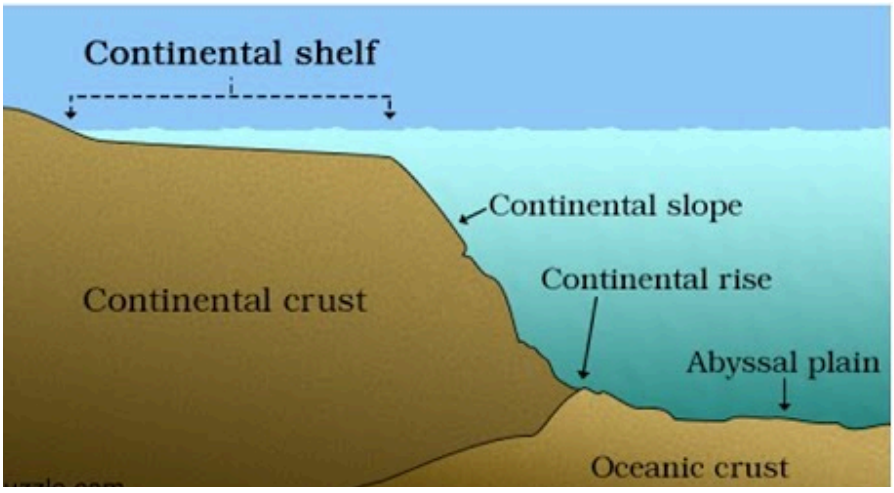
1. C
2. B
3. D
4. C
5. D
6. A
7. B
8. A
9. C
10. B
11. C
12. B
13. C
14. B
15. C
16. C
17. D
18. A
19. A
20. A
21. D
22. A
23. B
24. C
25. D
26. C
27. A
28. D
29. B
30. A

Open Ended Question Answer Key:

1. Natural disasters
    a. El Niño happens when there are sea surface anomalies greater than 0.5 deg. C across the central tropical Pacific ocean.
    b. El Niño Conditions are temporary signs that the tropical Pacific Ocean is starting to warm abnormally in a period of less than five months. An El Niño episode is a fully developed and sustained climate pattern characterized by significant and widespread warming of sea surface temperatures (SSTs) in the tropical Pacific which lasts for 5 months or longer.
    c. Warm tropical water replaces the cold water used to support populations of fish.
    d. El niño occurs when there is an abnormal amount of warm water, whereas la niña occurs when there is an abnormal amount of cold water.
    e. Upwelling is when cold, nutrient-rich water rises up to the surface to support marine ecosystems. El Niño weakens upwelling along eastern boundaries, while La Niña strengthens upwelling. El Niño causes phytoplankton decline and impacts zooplankton, fish, seabirds, and marine mammals, while La Niña increases plankton populations and increases the populations of these animal populations.
2. Boundary types
    a. Plates move away from each other. rift zones, mid-ocean ridges, and expanding ocean basins
    b. Pressure is reduced on the mantle below and magma flows out.
    c. Many answers apply. The most common answer should be mid-atlantic ridge.
    d. Rift-structure
    e. Ocean-water flows in and can cause rivers to form or affect bodies of water.
3. Waves
    a. Wave crest
    b. Wave trough
    c. One half the distance between the crest and the trough.
    d. Very small waves with very small wavelengths.
    e. Wind pressures on the water.
    f. Waves that move back and forth (oscillate) vertically. The wave appears like it’s moving forwards when it’s not.
4. Ocean sediments
    a. Lithogenous/terrigenous sediment
    b. Near continental margins
    c. Biogenous sediment
    d. Calcareous deposits from shells of marine organisms. Answers vary, though.
    e. Precipitating from the ocean water.
    f. Lithogenous and biogenous
    g. Cosmogenous sediment.
5. Surface currents
    a. Deflected to the left in the southern hemisphere or counter-clockwise direction
    b. Eastern boundary currents

c. Western boundary currents are faster, narrower, warmer and deeper than Eastern currents. Eastern boundary currents are relatively slower, broader, colder, and shallower than Western currents.
d. The Gulf Stream is the Western boundary current that flows north along the eastern coast of the US.
e. A gyre is classified as a big system of rotating ocean currents formed through the coriolis force, landmasses, and global wind circulation.
f. Surface currents help regulate marine ecosystems through nutrient cycling and weather patterns such as precipitation.

6. Geostrophic and ekman balances
    a. Differences in the water's height and therefore pressure across a horizontal region. Factors like variations in water density (due to temperature and salinity differences) create these pressure differences. This is because a higher density would result in a greater pressure gradient, whereas a smaller density results in a smaller pressure gradient. These pressure gradients determine the strength of a current.
    b. The pressure gradient always acts perpendicular to the isobars and from areas of high pressure to low pressure.
    c. The wind stress and coriolis effect produce the forty five degree angle in the ekman layer.
    d. The ekman balance is comprised of wind stress frictional forces and the coriolis force while the geostrophic balance is comprised of the pressure gradient force and the coriolis force.the ekman balance describes a smaller scale of water flow while geostrophic balances describe larger scale flows. Geostrophic balances have more effect in the deep ocean whereas ekman balances are more prevalent near the surface. Answers may vary.
7. Topographical features
    a. This is what the diagram should look like if filled in correctly:



    b. Continental shelf